

SAMSUNG

Product Experience Research

Using HTC Scribe on Flyer:
Pen-Tablet Product Experience
Evaluation

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Design & Technology Research

Strategic Marketing Research

Samsung Telecommunications America

The Value Proposition – Pen + Tablet

HTC Flyer & Scribe

HTC Scribe technology

HTC Flyer is the first tablet delivering a new HTC innovation that features natural pen interaction. It can draw, paint, write, sign documents, and even retouch pictures. It can be a highlighter for the reader (and by the way all your highlights will be collated when you're done). The HTC Flyer can even add voiceover to a storybook you're drawing up at home. Work or play, it's magic.



A tablet like no other



HTC Scribe technology



Supreme note taking



Streamlined unibody construction

The Value Proposition – Pen + Tablet

HTC Flyer & Scribe

Supreme note taking

HTC Flyer makes sure you capture everything with HTC Notes. You can add scribbles, photos and sound to your detailed notes. Link your notes to events in your calendar and your calendar transforms into a multimedia archive of your past meetings or events. And with Timemark, you can tap on any moment from your meeting and hear the audio you recorded, so you never have to struggle with deciphering the notes you took.



A tablet like no other



HTC Scribe technology



Supreme note taking



Streamlined unibody construction



HTC Scribe & Flyer Product Experience

Overall Results

Overview

This consumer behavioral and usability evaluation covered the use of HTC Scribe with HTC 7" Flyer Tablet.

Respondents (N=12) interacted with different applications on HTC Flyer tablet designed to use with HTC Scribe (a special pen as a differentiating input method).

Tasks

Without touching the pen or the tablet, respondents guessed the responses of the tablet when different buttons and user interface elements .

Respondents then performed typical tasks using pen/tablet combination (scribble, multimedia notes, annotation/highlighting of books (Reader) and document (Polaris Office)).

They rated the product experience and gave feedback when they completed the tasks.



Results

Interaction of Pen with Tablet

Most of the respondents were dissatisfied with the interaction design paradigm between pen and tablet . (slide 5 for summary)

Applications

Reactions to the pen-optimized applications were mixed. Respondents did not think the scribble and multimedia notes provided enough value. (Slides 16 -26)

Pen Palette and Options

Reactions to the Pen Palette and Options were generally very positive. This part of the product experience was well-designed and satisfying to most respondents. (Slides 27 to 37)

Product Concepts

Respondents did not think the product provided good value. The retail value of the tablet/pen was substantially higher than what consumers were willing to pay. (Slides 5, 42 to 46)

Research Insights

Seamless Integration of Pen and Touch

Seamless integration of pen and touch is critical to the success of tablet/pen combination. Respondents expected the pen to function as their finger extension of touch with writing/drawing and hand-writing recognition capabilities.

Design Recommendations

1. A button can be used on the pen to enter the "draw" mode by pressing and holding the button. The button should be placed at a natural position and can be activated almost effortlessly by holding the pen. With a slight change in the way the pen is held, users can alternate between using the pen as a stylus and as a drawing device.
2. Pen user experience = finger touch + writing/reading. (Slide 6)

Limitations

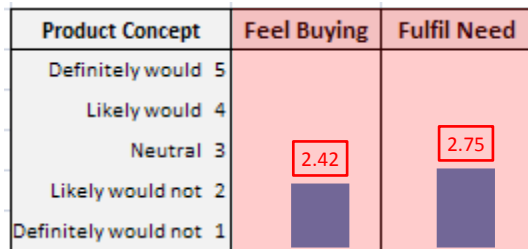
Although the respondents in the research were smartphone users (either tablet owners or intenders) , they were not screened for interest or need in using a specially-designed pen with tablet.

Results from Consumer Behavioral Research

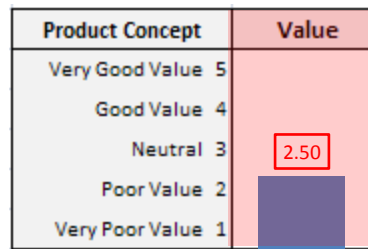
HTC Flyer & Scribe

Insight:
HTC Flyer & Scribe fail to deliver the value proposition because of major interaction design issues between pen and touch input. Half-baked products can leave users frustrated, creating backlash against the products and the brands.

Product Concept



Product Value

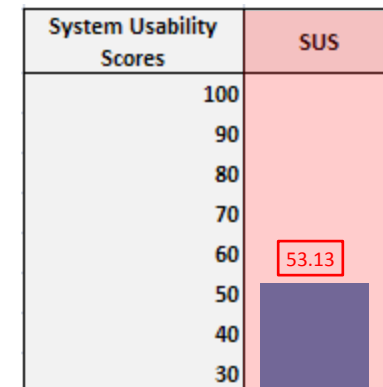


**“I want to Put a Block of C4 on this thing.
 Totally Pointless”**
 written comment from a respondent dated July 6, 2011.

Overall Usability



System Usability Scores



Research Insight from Consumer Behavioral Research

Design Recommendations

Insight:

The pen should function as a stylus (alias finger extension). A dedicated button should be used to activate the pen function (as writing/drawing instrument). No dedicated button for the pen on the device is necessary.

Physical Design

1. The pen should function as a stylus (alias finger extension) by default.
2. A dedicated button should be used to activate the pen function (as writing/drawing instrument). Writing and drawing can be accomplished by holding the pen naturally with the button pressed.
3. An alternative method is to define a user-initiated method of entering and leaving the writing/drawing mode (e.g., by a specific gestures or action with the pen like double tap)
4. No dedicated pen button on the tablet is necessary.

User Experience Design

1. Change of mode of the pen function between writing/drawing and pointing/selecting instrument should be designed as the same input toggle gesture with both pen or finger (either holding a dedicated button or using the same gesture). A “mode” indicator may be useful in the status/notification area.
2. It is better to add additional pen input support for existing apps than creating completely new apps trying to replicate.
3. Any actions in any apps that can be invoked by touch should be supported by pen by default (and vice versa). Users should be able to draw and write with finger.

Application Design

1. Although some cross-platform capability is supported (e.g., Word in Polaris Office), there are some minor kinks and issues that need to be addressed. Consumer education will be needed here as there are skepticisms of cross-platform compatibility.
2. Handwriting recognition capability is expected from consumers.

Session Protocol

METHODOLOGY

Synopsis of Research

HTC Flyer/Scribe Product Experience

Goals

1. To evaluate the product experience of HTC Flyer and HTC Scribe as a unique and differentiating feature from HTC, focusing on tablet/pen interaction design and use cases.
2. To uncover design and usability problems of tablet/pen combination products and use the learning to improve similar Samsung products.

Research Sessions

The sessions were conducted by STA Design & Technology Product/User Research in San Diego between July 5 and July 8, 2011. The respondents (N=12) were extra backup respondents for tablet form factor research.

Recruiting Criteria

Tablet users and tablet intenders (All smartphone users)



HTC Flyer/Scribe Product Experience Evaluation (tablet-pen interactions)

Methodology: Session Protocol

User Expectations

- Respondents described typical scenarios and expectations of how they would use the products if they were given a tablet and a pen that would enable them to draw on the tablet.
- Respondents were then shown the products without using them or any demonstrations of their behavior.
- Respondents guessed the functions of the 2 buttons on the pen without using it.
- Respondents guessed the responses of the tablet user interface pretending that they were tapping the pen in different areas of screen and the buttons.

Product Experience Evaluation

- Respondents verified their guesses by using the pen on the tablet.
- Respondents evaluated usability of 5 different tasks/use cases using apps designed specially for Scribe Scribble, Notes, Reader and Polaris Office in the tablet.
- Respondents evaluated usability of Pen Options Palette (6 different menu)
- After evaluating the above tablet-pen usage scenarios, respondents completed an overall evaluation of tablet/pen combination.

Product Concept Evaluation

- Respondents completed the product concept and pricing evaluation.



Pen & Tablet Experience

USER EXPECTATIONS & ASPIRATIONS

Consumer Expectations – Pen/Tablet Products

Features/Functions Expectations of Pen/Tablets

Insight:
Consumers expect the pen supports the same functions as finger touch and more.. as a drawing, writing instrument capable of hand-writing recognition..

Respondents' Expectations of how they use the Pen on the Tablet



Pen & Tablet Experience

TABLET-PEN DESIGN EVALUATION

HTC Scribe – Pen Design & Functions

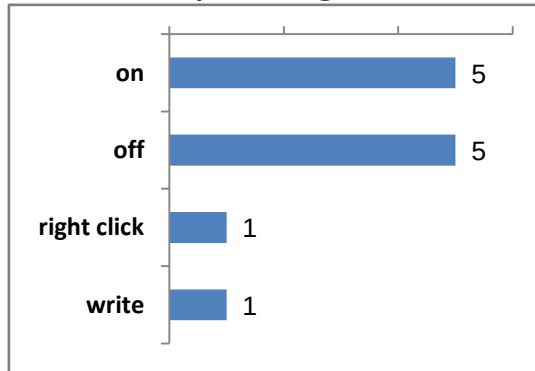
Guessing the Function Buttons on the Pen

Insight:
The function buttons of HTC Scribe are unlikely to be useful for consumers as designed.

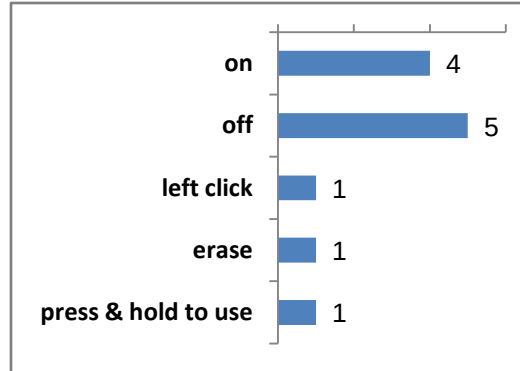
a None of the respondents made the correct guess.

b None of the respondents made the correct guess.

Top button guesses



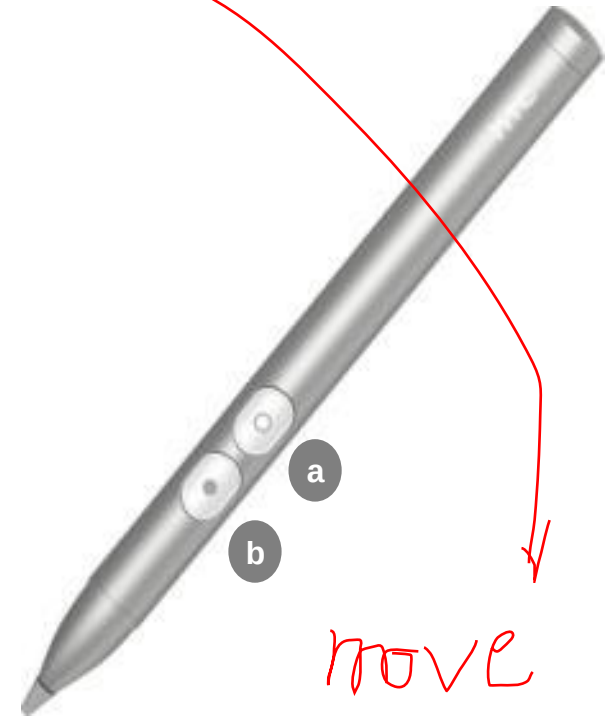
Bottom button guesses



N=12 Guess the functions of the two buttons on the pen by looking at the icons.

User Interface Elements		Actual Behavior	Error rate
Top Button	a	Eraser	12/12
Bottom Button	b	Highlighter	12/12

Unacceptably high error rate 12/12



HTC Scribe – Pen Design & Functions

Guessing the Behavior of Tablet with Pen tapping on different areas

Insight:

The user experience of the tablet/pen combination are highly non-intuitive and error prone. The interaction design fails to match the mental model of most consumers. They assume the pen will function the same as touch.

All respondents did not guess correctly from area “a” through “l” (except area “d”).

d 11/12 made the correct guess that the button was related to pen.

N=12

Guess what will happen when one taps the pen on the different areas of the homescreen and the 4 buttons.

User Interface Elements		Actual Behavior	Error rate
Home Button	a	Nil	12/12
Menu Button	b	Nil	12/12
Back Button	c	Nil	12/12
Pen Button	d	Pen Options	1/12
Widget (Time)	e	Take Snapshot (Scribble)	12/12
App Shortcut (Market)	f	Take snapshot (Scribble)	12/12
Launcher Bar	g	Take snapshot (Scribble)	12/12
Homescreen Background	h	Take snapshot (Scribble)	12/12
Notification Bar	i	Take snapshot (Scribble)	12/12

Unacceptably high error rate 12/12

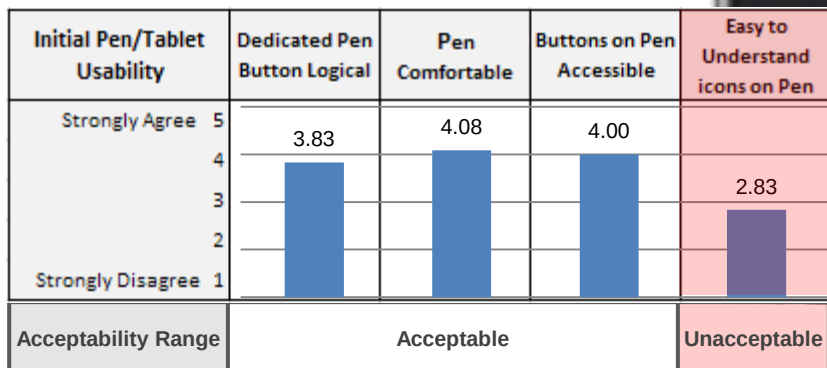


HTC Scribe/Flyer – Initial Pen/Tablet Product Experience

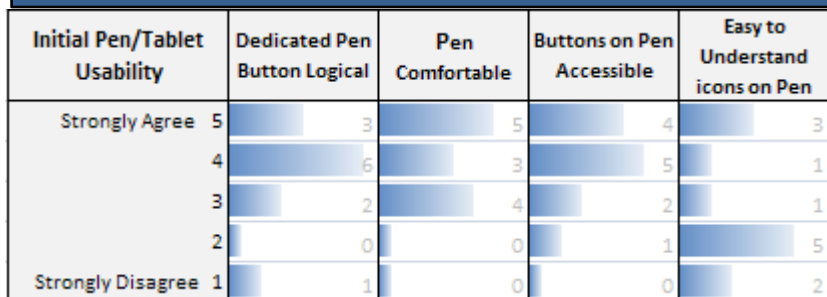
Initial Impressions Ratings

Insight:
Initial impressions of tablet/pen combination are generally positive except the icons on the buttons of the pen. However, 7 out of 12 respondents expressed negative remarks ~~(from mild to extreme)~~ about using the pen as designed.

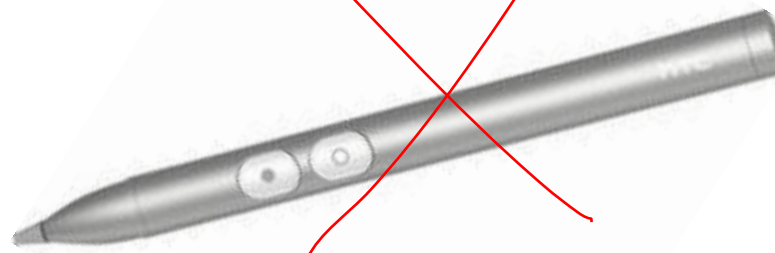
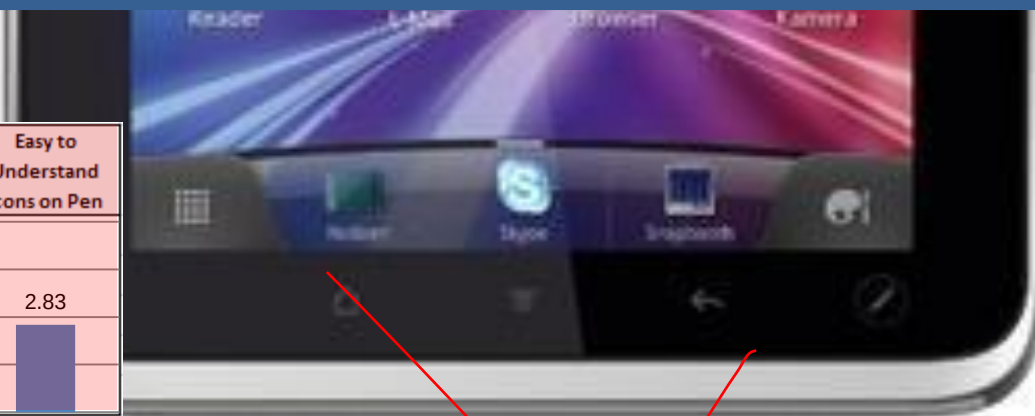
Initial Impressions of Tablet/Pen Combination



Icons on the buttons on the pen were confusing to respondents.



Distribution of number of respondents (total N=12)



“Pen felt light-weight and cheap”

written comment from a respondent dated July 6, 2011.

Scribble: Home Page Snapshot

Scribble: Browser Page Snapshot

Using and Sharing Multimedia Annotations with Notes

Annotations and Highlighting in Reader

Annotations and Highlighting in Polaris Office (Word document)

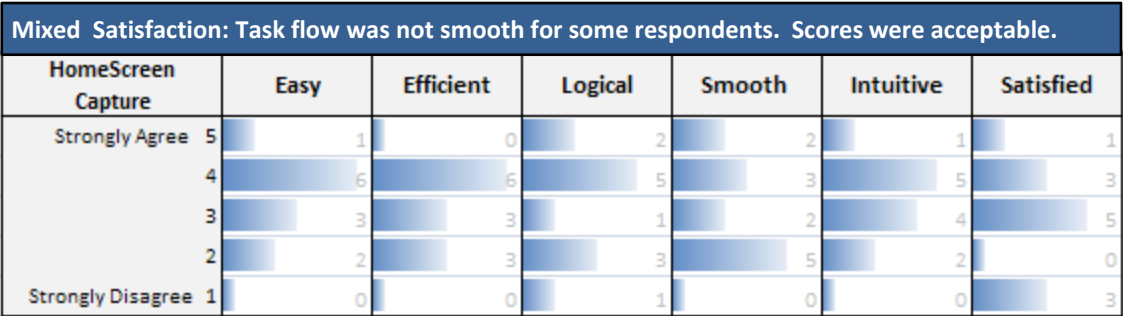
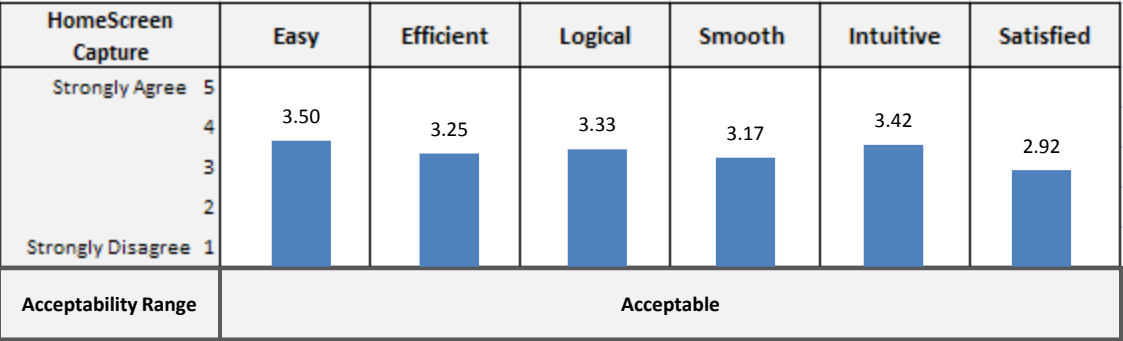
USE CASES: USABILITY EVALUATION

HTC Scribe/Flyer – Scribble

Usability & Satisfaction Rating: HomeScreen Capture/snapshot with free-hand pen edit

Insight:
Consumers find it difficult to anticipate when and where in the task to use pen and when/where in the task to use touch.

Task: Homescreen capture screenshot and share the picture with navikey3@gmail.com via Gmail.



Distribution of number of respondents (total N=12)

not critical



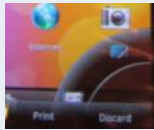
HTC Scribe/Flyer – Scribble

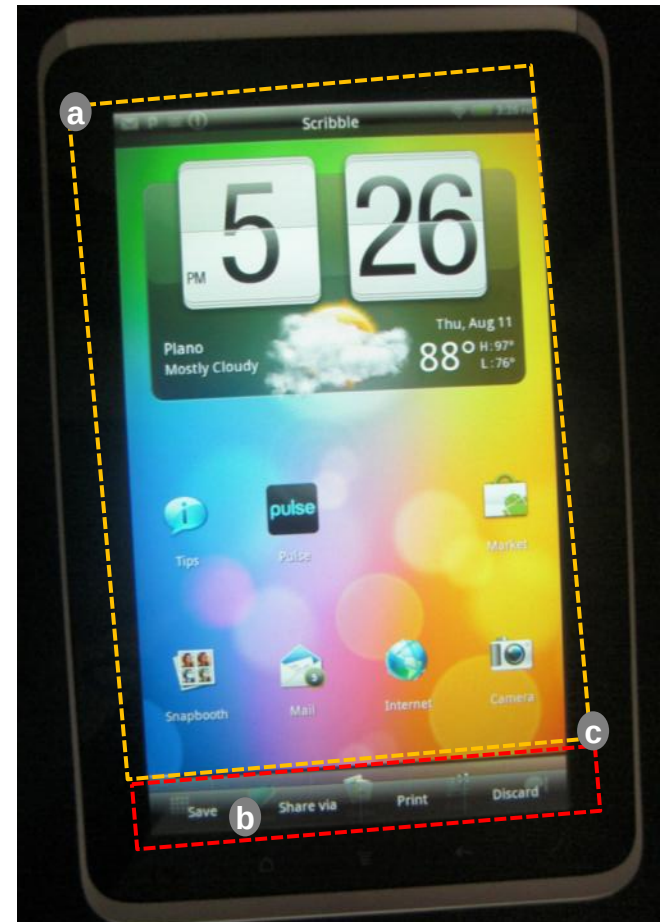
Usability Issues: HomeScreen Capture/snapshot with free-hand pen edit

Insight:

Consumers do not anticipate capturing screen with a pen and edit the image in everyday use.

ng

	Observations	Comments
Context of Use	Respondents questioned the utility of using the pen for screen capture rather than using it as a stylus. They also did not anticipate much use of screen capture in everyday activities.	Screen capture did not have everyday utility to support the exclusive use of the pen for snapshot.
Interaction Design (b)	Respondents repeatedly used pen to select options "Save", "Share via", "Print", "Discard", despite they knew pen would not work on those options. Only 1 respondent out of 12 thought the process was reasonable.	Design did not match mental model of usage: the behavior of pen and finger tap confused respondents after repeated use but they would correct errors quickly by swapping between pen and finger.
Visual Design (a) (b)	Most respondents did not notice the subtle changes with the screen capture rendered in scribble. They also did not notice the title of the app ("scribble") options appeared (and disappeared) when they tapped the screen with their fingers.	The full screen display of the captured image was particularly confusing as the image looked the same as the actual home (or app) screen.
Capability & Information Architecture (c)	One respondent suggested that the presentation of the Pen Options palette immediately after the screen capture would help to signal the transition. ✓	Bringing options upfront reduced confusion. 



HTC Scribe/Flyer – Scribble: Browser Screen Snapshot

Usability & Satisfaction Rating

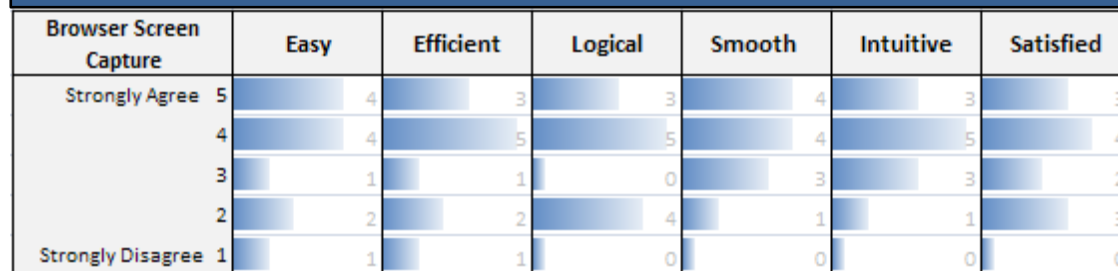
Insight:

Consumers think this can be done by sending the link to the web page and describing the message they want to convey to recipients through email.

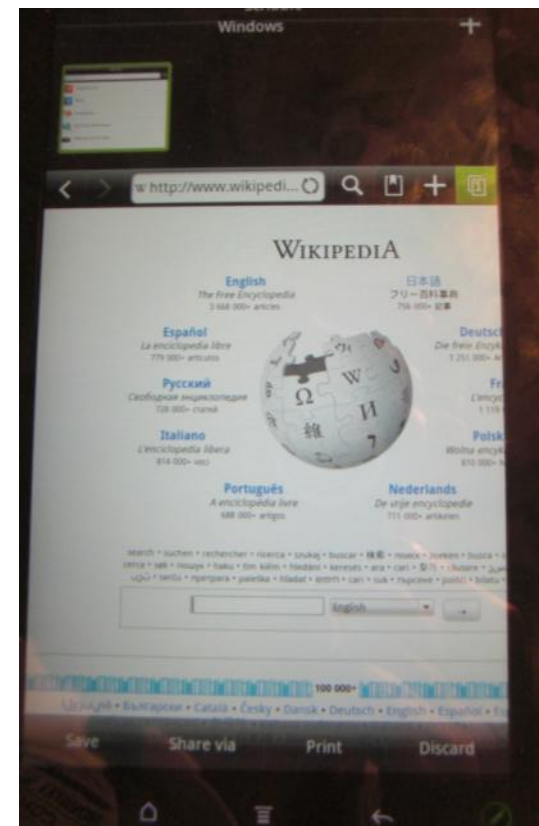
Task: Browser page snapshot: capture Wikipedia.org and discard the capture



Mixed Satisfaction: User interface was not logical for some respondents. Scores were generally acceptable.



Distribution of number of respondents (total N=12)

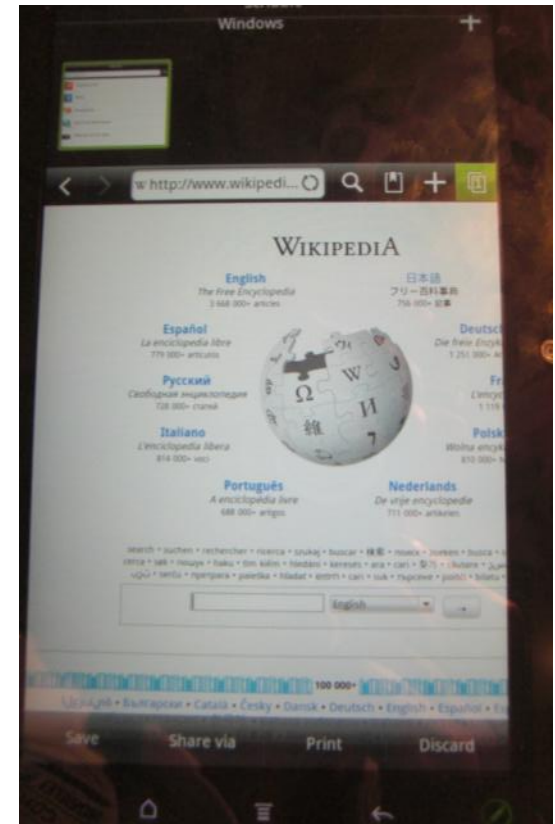


HTC Scribe/Flyer – Scribble: Browser Screen Snapshot

Usability Issues: Browser Capture/snapshot with free-hand pen edit

Insight:
Consumers find it difficult to anticipate when and where in the task to use pen and when/where in the task to use touch.

	Observations	Comments
Context of Use	One respondent said that the web page could be sent through email and any comments can be added in the email rather than using this app.	This function/feature may be more useful for web site designers and tablet developers than consumers.
Interaction Design	Five respondents remarked <u>explicitly</u> again using pen and finger touch should behave the same. Two respondents thought it was easy <u>to</u> do once they figured it out.	Design did not match mental model of usage: same issue as with homescreen capture.
Visual Design	<u>Same as Homescreen capture.</u> <i>not clear</i>	<u>Same as Homescreen capture.</u>
Capability & Information Architecture	One respondent wanted to take “snapshot” of the entire web page (both visible and hidden portion of the page)	The intended capture experience should be considered in relationship to the actual content of the web page, rather than an image.

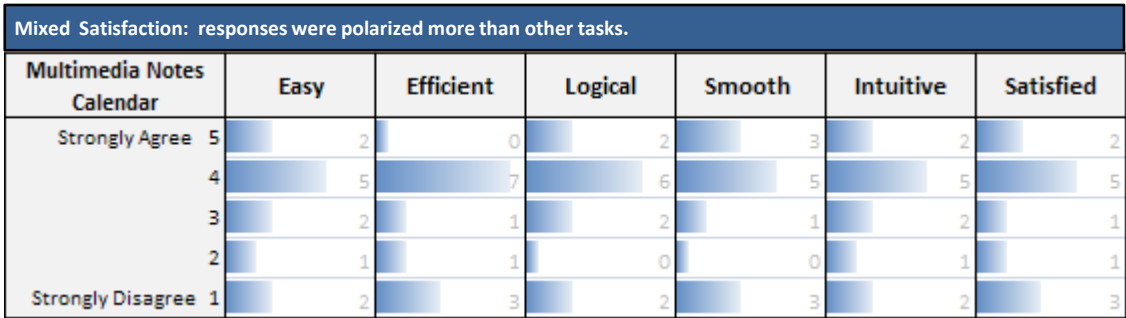
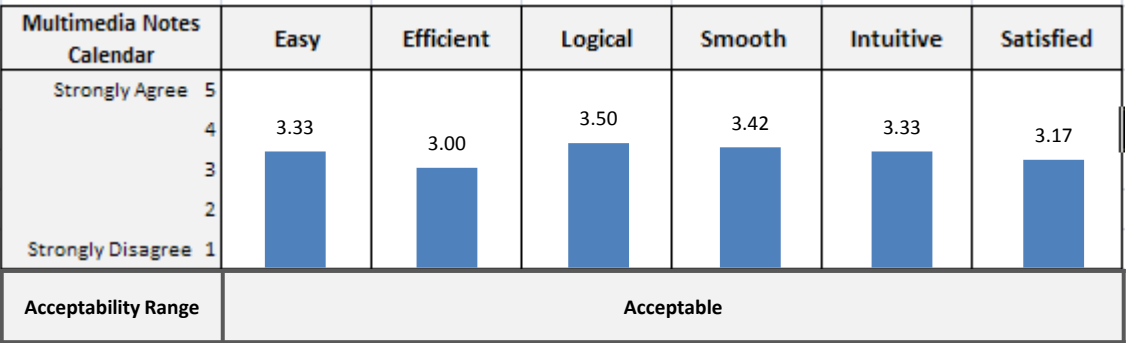


HTC Scribe/Flyer – Multimedia Notes

Usability & Satisfaction Rating: Maps, Notes, Voice Memo, Calendar and Share

Insight:
Consumers are not likely to have major issues with multimedia notes, but they think they do not need such feature-rich multimedia apps to accomplish what they need.

Task: Maps: Capture screenshot and save (to Notes), annotate the map using different pen types, strokes and colors, record a brief message when you are annotating and associate it with the calendar and share the notes with navikey3@gmail.com



Distribution of number of respondents (total N=12)



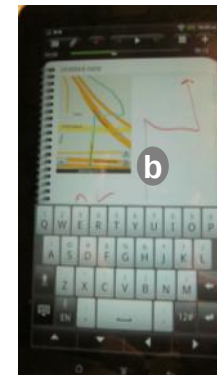
HTC Scribe/Flyer – Multimedia Notes *task not clear*

Usability Issues: Maps, Notes, Voice Memo, Calendar and Share

Insight:

Multimedia notes cannot zoom in/out captured images. Consumers have to consciously zoom the map to the correct size before image capture. In order to develop a really useful application, pen edits have to work with each application seamlessly (e.g., within Google Maps).

	Observations	Comments
Context of Use a	Five respondents observed that this application was cumbersome, confusing or inefficient after using it. Only 1 respondent had positive remarks.	The integration of multimedia (voice, scribble, text, images, videos..) and calendar fails to impress consumers after they used the application.
Interaction Design b	The complexities of dealing with different media, coupled with the problem between using pen or finger made this application complicated and tedious.	Although all the integrated features are available at fingertips, the over-designed features make the experience cumbersome.
Visual Design c	The screen layout was busy with a spiral notebook, options for playback, forward and rewind (without anything to playback).	The layout is cluttered with unnecessary “spiral notebook” metaphor representation and irrelevant options (disabled) in the context of use.
Capability & Information Architecture d e	Two respondents wanted to resize the map after capturing it into multimedia notes but could not.	The complexities of this application can be managed by layering features and associated options according to usage priority.

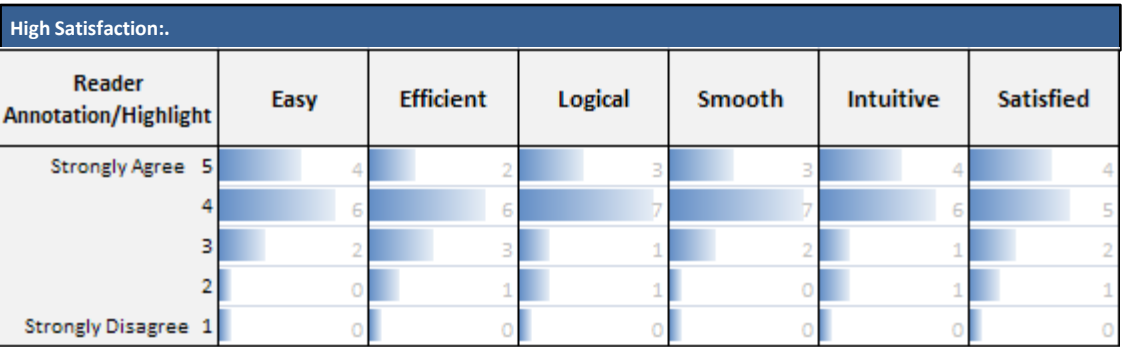
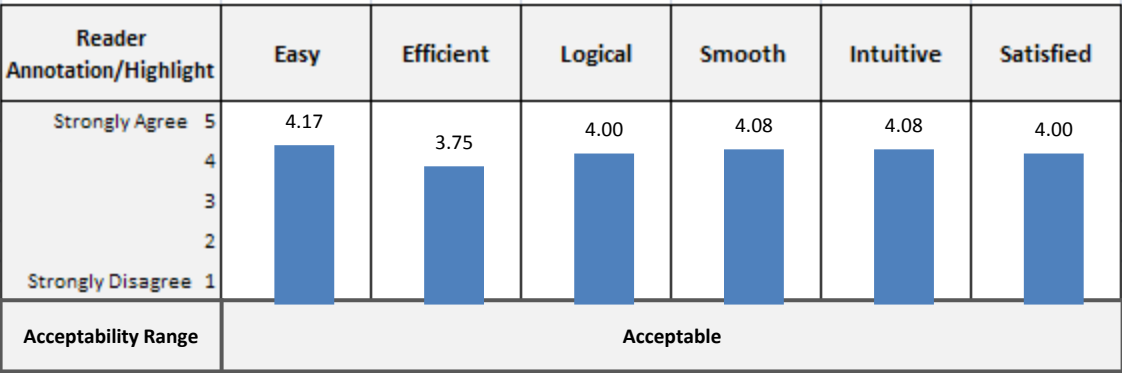


HTC Scribe/Flyer – Reader

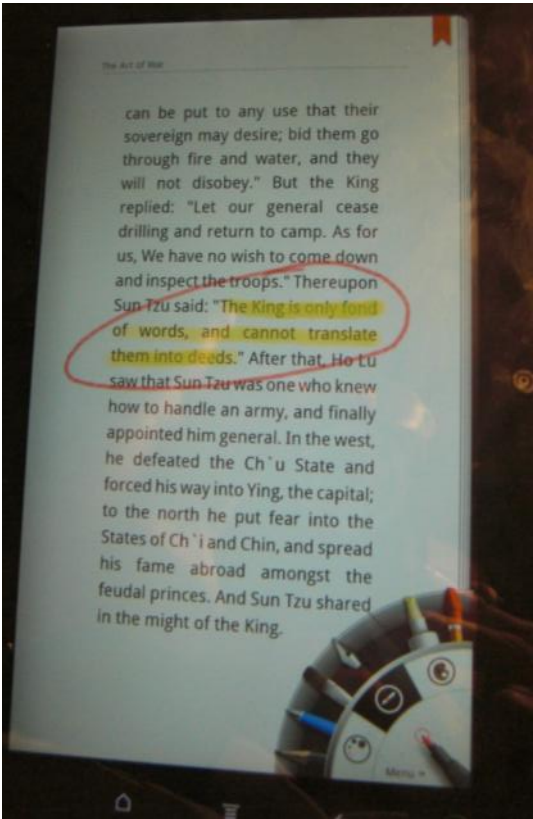
Usability & Satisfaction Rating: Reader Annotation & Highlight

Insight:
Consumers find annotation and highlighting in Reader easy and somewhat useful. They are generally satisfied with using the pen for this use case.

Task: Reader: annotate/highlight the text using different pen types, strokes and colors.




 Distribution of number of respondents (total N=12)

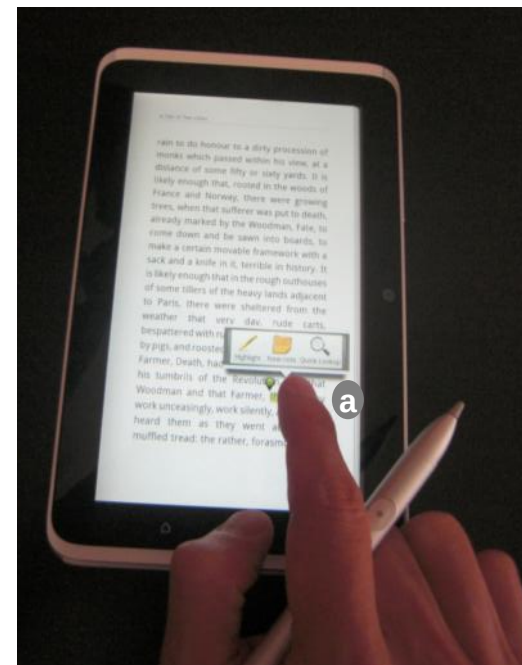


HTC Scribe/Flyer – Reader

Usability Issues: Reader Annotation & Highlight

Insight:
The inability to use pen to undo highlight or lookup a word create problems for consumers. Although design experience replicates the natural habit of using writing instrument for annotation and hand/finger for flipping pages, this design metaphor does not necessarily meet users' changing expectations. Consumers would assume the pen as an extension of finger touch and can be used to accomplish same functions as touch (flipping pages).

	Observations	Comments
Context of Use	Three respondents thought the application was easy and useful. One respondent did not see any need for using the pen to annotate.	Users should be able to annotate and highlight with a well-designed reader app.
Interaction Design	Respondents expected the capability of “undoing” highlighting using the pen (without using finger touch). To undo an operation like highlight should be supported by similar input methods.	To look up a word or insert a new note, one must use finger touch by touching on the word. Users should be able to accomplish any functions within the app by either pen and finger touch input.
Visual Design	Respondents were not aware of using finger touch to “undo” highlight (and choices of highlight color).	A consistent representation of objects with menus accessible by touch and pen .
Capability & Information Architecture	One respondent desired more “editing” features with the app.	Options and functions accessible by touch should also be accessible by pen .



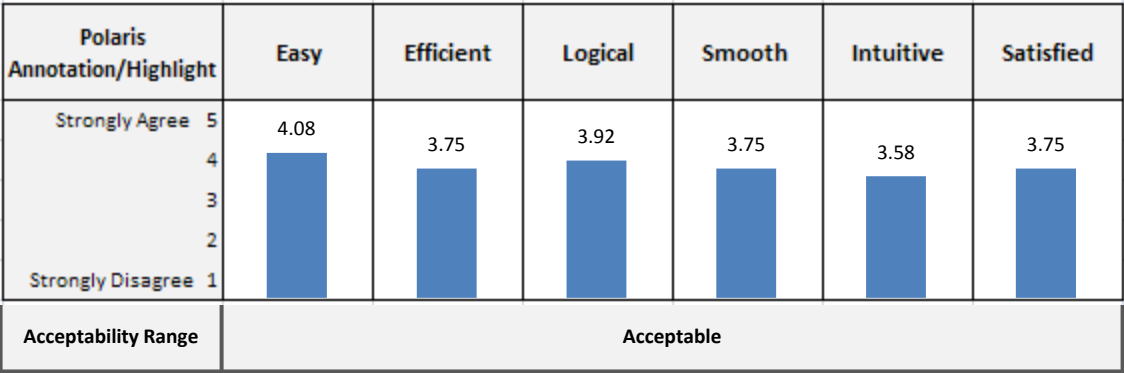
HTC Scribe/Flyer – Polaris Office

Usability & Satisfaction Rating: Word document Annotation & Highlight

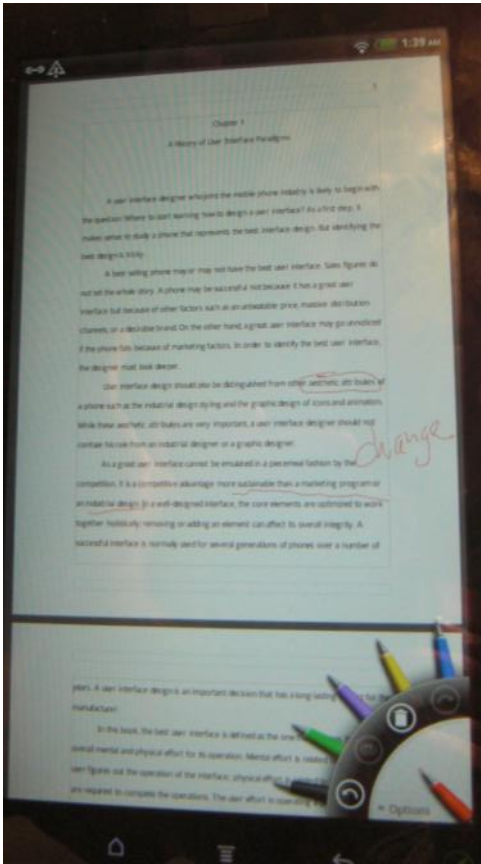
Insight:

With the touch input, users expect the pen as their natural extension of the finger. Separating finger and pen input as two different input categories with different functions violates users expectations and creates confusion.

Task: Polaris Office (Word): annotate & highlight the text using different pen types, strokes and colors.



Distribution of number of respondents (total N=12)

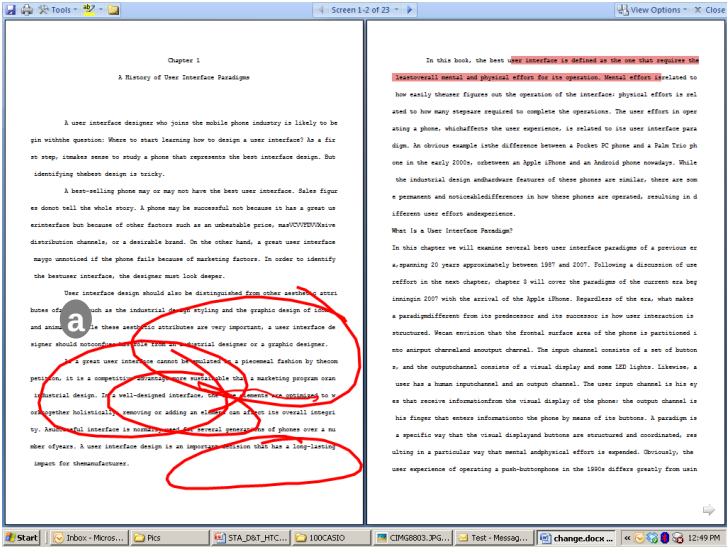


Usability & Capability Issues: Word document Annotation & Highlight

Insight:

Some consumers are concerned with cross-platform compatibility of adding annotations and highlight (note: this is supported but the “highlight” is not the same as “highlight” in Word. Background fill is used instead).

	Observations	Comments
Context of Use	Two respondents were concerned about cross platform compatibility.	Users should be able to annotate and highlight with a well-designed reader app.
Interaction Design	One respondent remarked that it was not as intuitive and easy as the Reader app. Another respondent mentioned that once it was learned, it was OK. One felt that it was good and easy.	The pen/touch dichotomy was noted by two respondents. One suggested to integrate touch with a “third” button on the pen.
Visual Design	Respondents were not aware of using finger touch to “undo” highlight (and choices of highlight color).	A consistent representation of objects with menus accessible by touch and pen .
Capability & Information Architecture	One respondent desired more “editing” options while another one wanted copy & paste with pen. One respondent wanted XML-based pen edits for cross platform compatibility.	Cross platform compatibility is already supported. User education will help to address those concerns.



Scribble/Notes

Hiding/Unhiding Pen Strokes, Undo/Redo, Clear All Pen Strokes

Changing Size of Pen Strokes

Changing Types of Pen

Changing Color of Pen Strokes

PEN OPTIONS PALETTE: USABILITY EVALUATION

HTC Scribe/Flyer – Pen Options Palette: Scribble & Notes

Usability & Satisfaction Rating

Insight:
 The association between app icons (without textual labels) and apps can be difficult for some consumers, especially when the applications overlap in features and functions like the Notes and Scribble applications in the Flyer.

Action: create Scribble (Snapshot) or Notes

Pen Options Scribble/Notes	Easy	Efficient	Logical	Smooth	Intuitive	Satisfied
Strongly Agree 5	4.00	3.83	3.83	3.67	3.58	3.75
4						
3						
2						
Strongly Disagree 1						
Acceptability Range	Acceptable					

Moderate Satisfaction:						
Pen Options Scribble/Notes	Easy	Efficient	Logical	Smooth	Intuitive	Satisfied
Strongly Agree 5	3	2	1	1	2	2
4	7	6	9	8	5	7
3	1	4	1	1	3	1
2	1	0	1	2	2	2
Strongly Disagree 1	0	0	0	0	0	0


 Distribution of number of respondents (total N=12)

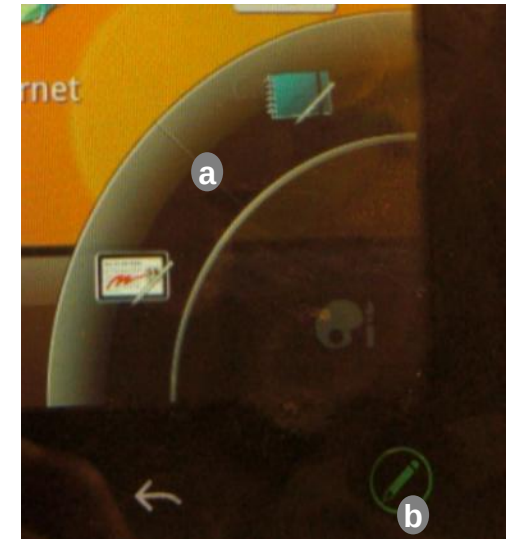


HTC Scribe/Flyer – Pen Options Palette: Scribble & Notes

Usability Issues: Pen Options Palette

Insight:
Some consumers think that tapping on the dedicated “pen” button using the pen as the only way to invoke the pen options palette unnecessary and confusing.

	Observations	Comments
Context of Use	One respondent felt that Scribbles and Notes served similar functions. Another wanted extra options to use “Cloud” services for captured image upfront.	The apps highlighted in the Pen Options Palette did not address users’ need adequately.
Interaction Design a	One respondent thought it required extra effort to invoke Pen Options Menu by tapping on dedicated pen button on tablet and suggested using a button on pen to invoke in-place pop-up menu. Another wanted upfront pen-selection.	The dedicated pen button on the tablet is more of a nuisance as respondents have to go out of the way to tap on it for Pen Options Palette.
Visual Design b	Two respondents desired immediate presentation of pen options menu in Scribbles.	Application Icons without textual labels can be difficult to associate with the apps.
Capability & Information Architecture a	One respondent thought the options were too limited. Three respondents desired more functions/features in Pen Options Palette.	Feature creeping – current features should work.



HTC Scribe/Flyer – Pen Options Palette: Hide/Unhide, Clear pen Strokes, Undo/Redo

Usability & Satisfaction Ratings

Insight:

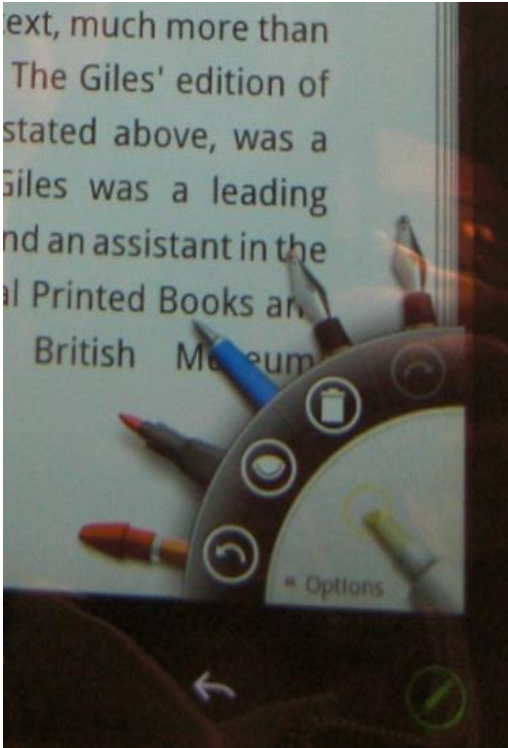
Consumers have no problems with the Pen Options Palette and are generally satisfied with the user experience with these options/functions.

Action: Hiding/unhiding pen strokes, undo/redo, clear all pen strokes

Pen Options (Un)Hide,(Un/Re)do		Easy	Efficient	Logical	Smooth	Intuitive	Satisfied
Strongly Agree	5	4.25	4.33	4.00	4.25	4.17	4.17
	4						
	3						
	2						
Strongly Disagree	1						
Acceptability Range		Highly Acceptable					

High Satisfaction:							
Pen Options (Un)Hide,(Un/Re)do		Easy		Efficient		Logical	
Strongly Agree	5		5		5		3
	4		5		6		6
	3		2		1		3
	2		0		0		0
Strongly Disagree	1		0		0		0


 Distribution of number of respondents (total N=12)



HTC Scribe/Flyer – Pen Options Palette: Hide/Unhide, Clear pen Strokes, Undo/Redo

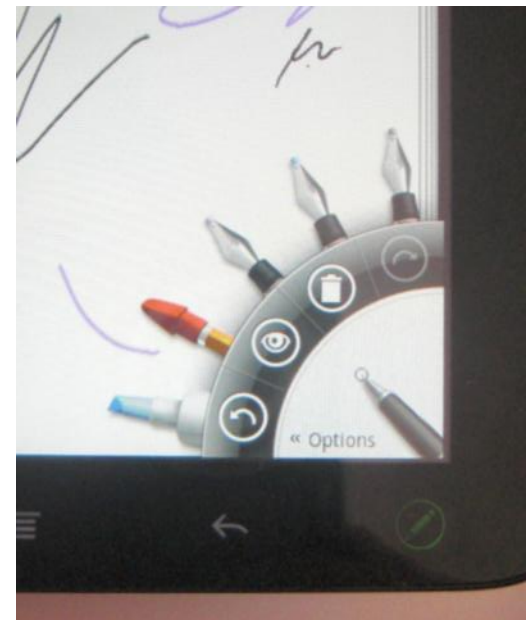
Usability Issues

Insight:

Consumers do not have any major problems with the experience. *of?*

not an insight

	Observations	Comments
Context of Use	No comments from respondents	The current design seems to work out well for most respondents.
Interaction Design	No comments from respondents	The current design seems to work out well for most respondents.
Visual Design	One respondent complained of the small icons used in the Options Palette.	Larger menu can be considered.
Capability & Information Architecture	No comments from respondents	The current design seems to work out well for most respondents.



HTC Scribe/Flyer – Pen Options Palette: Changing size of pen strokes

Usability & Satisfaction Ratings

Insight:
Consumers do not have major problems and are generally satisfied with the design.

Action: Changing size of pen strokes

Pen Options Changing Size	Easy	Efficient	Logical	Smooth	Intuitive	Satisfied
Strongly Agree 5	4.33	4.42	4.33	4.33	4.17	4.25
4						
3						
2						
Strongly Disagree 1						
Acceptability Range	Highly Acceptable					

High Satisfaction:						
Pen Options Changing Size	Easy	Efficient	Logical	Smooth	Intuitive	Satisfied
Strongly Agree 5	6	6	5	5	4	4
4	4	5	6	6	6	7
3	2	1	1	1	2	1
2	0	0	0	0	0	0
Strongly Disagree 1	0	0	0	0	0	0

Distribution of number of respondents (total N=12)



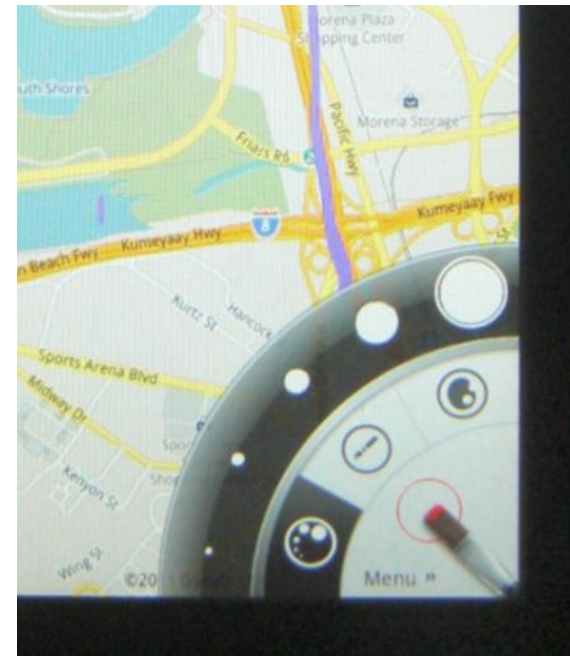
HTC Scribe/Flyer – Pen Options Palette: Changing size of pen strokes

Usability Issues

Insight:

Although some consumers may ask for more features, the current design should be adequate for common use cases.

	Observations	Comments
Context of Use	Most respondents did not have negative comments on the usage. Two thought that it was well-designed, very easy and self-explanatory	The current design seems to work out well for most respondents.
Interaction Design	One respondent wanted customizable mix and match.	Mix and match may complicate the user experience. The current design seems to work out well for most respondents.
Visual Design	No comments from respondents	The current design seems to work out well for most respondents.
Capability & Information Architecture	One respondent thought the options were too limited (customizable mix and match was desired) but another one thought it was too much.	The current design seems to work out well for most respondents.



HTC Scribe/Flyer – Pen Options Palette: Changing types of pen

Usability & Satisfaction Ratings

Insight:

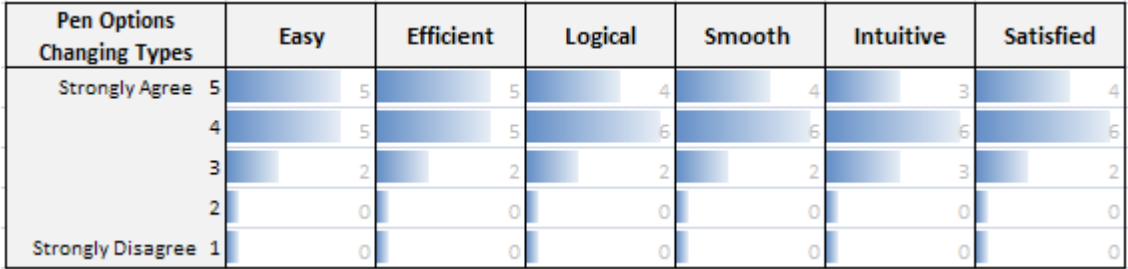
Consumers are generally satisfied with this function.

Action: Changing types of pen

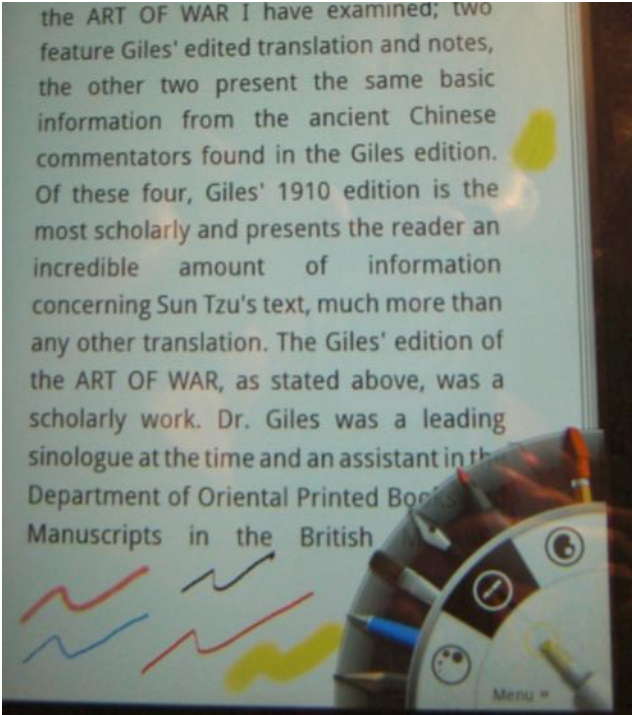


Acceptability Range	Acceptable
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Moderate Satisfaction:



Distribution of number of respondents (total N=12)

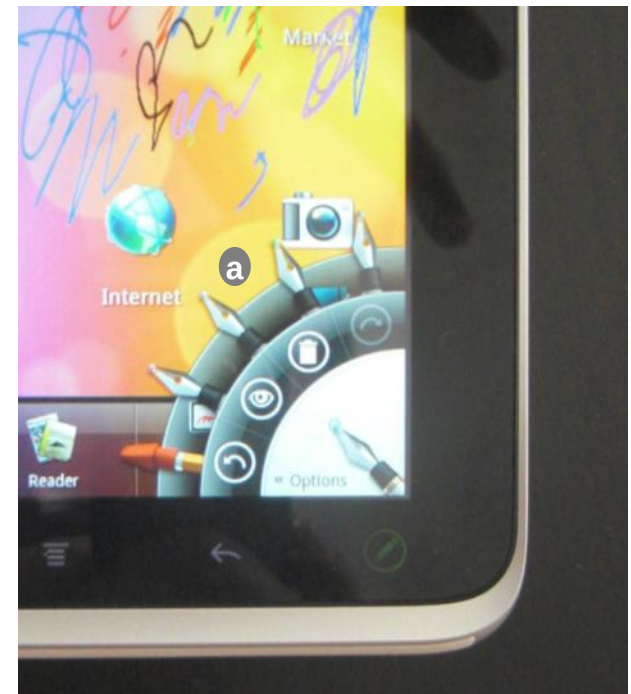


HTC Scribe/Flyer – Pen Options Palette: Changing types of pen

Usability Issues

Insight:
One visual design problem observed is the choice of different color pens does not show the colors well (the color patch associated with each pen is too small).

	Observations	Comments
Context of Use	Most respondents did not have negative comments on the usage. One thought that it worked well.	The current design seems to work out well for most respondents.
Interaction Design	No comments from respondents	The current design seems to work out well for most respondents.
Visual Design a	One respondent remarked that the icon to change pen was not obvious, but once found, it worked out well.	The color associated with each pen should be made larger (e.g., by using bands of color on their handles).
Capability & Information Architecture	No comments from respondents	The current design seems to work out well for most respondents.



HTC Scribe/Flyer – Pen Options Palette: Changing colors of pen strokes

Usability & Satisfaction Ratings

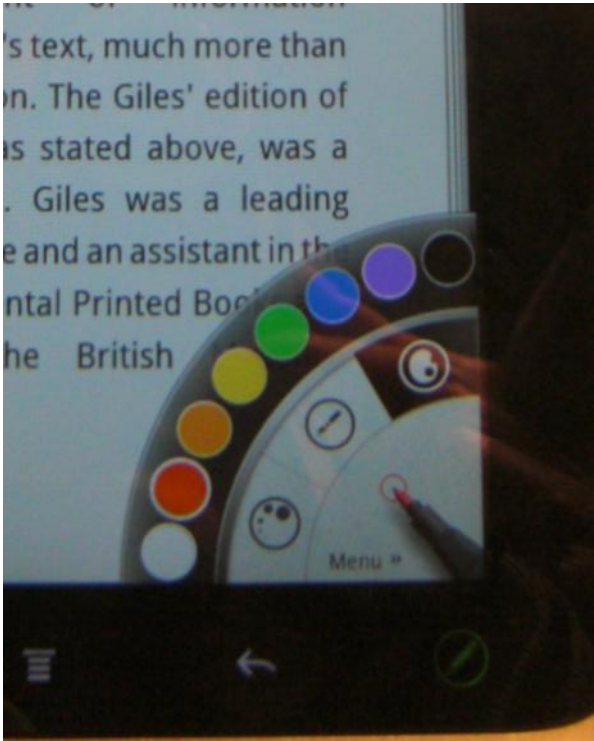
Insight:
Consumers are generally satisfied with the user experience of changing color palette. The capability to create a custom color can be added.

Action: Changing colors of pen strokes

Pen Options Changing Colors	Easy	Efficient	Logical	Smooth	Intuitive
Strongly Agree 5	4.25	4.33	4.25	4.17	4.17
4					
3					
2					
Strongly Disagree 1					
Acceptability Range	Acceptable				

Moderate Satisfaction:					
Pen Options Changing Colors	Easy	Efficient	Logical	Smooth	Intuitive
Strongly Agree 5	5	5	4	4	5
4	5	6	7	6	4
3	2	1	1	2	3
2	0	0	0	0	0
Strongly Disagree 1	0	0	0	0	0

Distribution of number of respondents (total N=12)



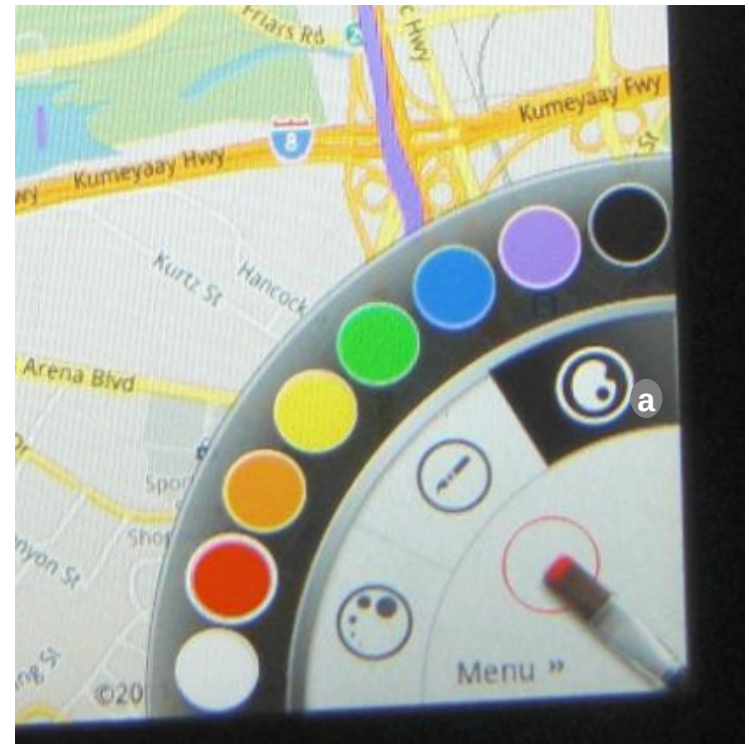
HTC Scribe/Flyer – Pen Options Palette: Changing color of pen strokes

Usability Issues

Insight:

The interaction is generally well-designed. One worthwhile feature to pursue may be the capability to add custom color for professional artists and designers as a niche product for those segments.

	Observations	Comments
Context of Use	Most respondents did not have negative comments on the usage. Two thought that it worked well and/or self-explanatory	The current design seems to work out well for most respondents.
Interaction Design	One desired the capability of mixing colors.	Adding and/or Mixing color can be designed in a deeper menu hierarchy. The current design seems to work out well for most respondents.
Visual Design a	One respondent wanted bigger icons while another one more meaningful icons.	The color palette icon can be improved by adding tints of color. The current design seems to work out well for most respondents.
Capability & Information Architecture	One respondent wanted the capability of mixing palette colors. Another one wanted more color choices.	Adding custom color can be designed in a deeper menu hierarchy. The current design seems to work out well for most respondents.



Pen

Tablet & Pen Combination

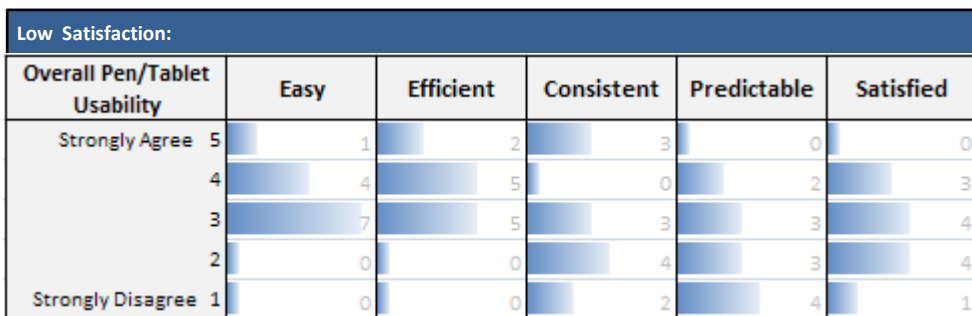
OVERALL USABILITY EVALUATION

HTC Scribe/Flyer – Overall Pen/Tablet Product Experience

Usability & Satisfaction Ratings

Insight:
Consumers find HTC Flyer/Pen product experience inconsistent across applications, difficult to predict where and when finger tap and pen tap to be used within and between applications and dissatisfied with the experience.

Overall Usability



Distribution of number of respondents (total N=12)

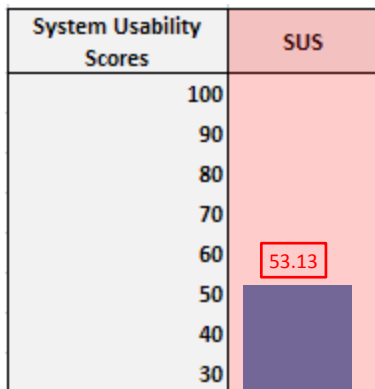


HTC Scribe/Flyer – Overall Pen/Tablet Product Experience

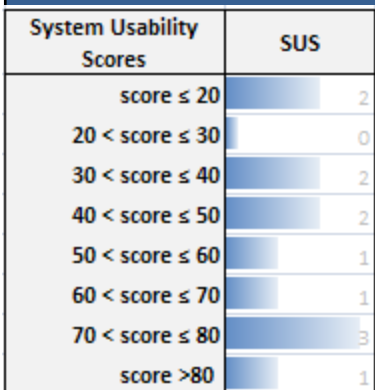
Usability & Satisfaction Ratings

Insight:
The System Usability Scores for overall Pen/Tablet Interactions were low and characteristic of major usability & design problems for consumers.

System Usability Scores



Wide distribution of SUS Scores



Distribution of number of respondents (total N=12)



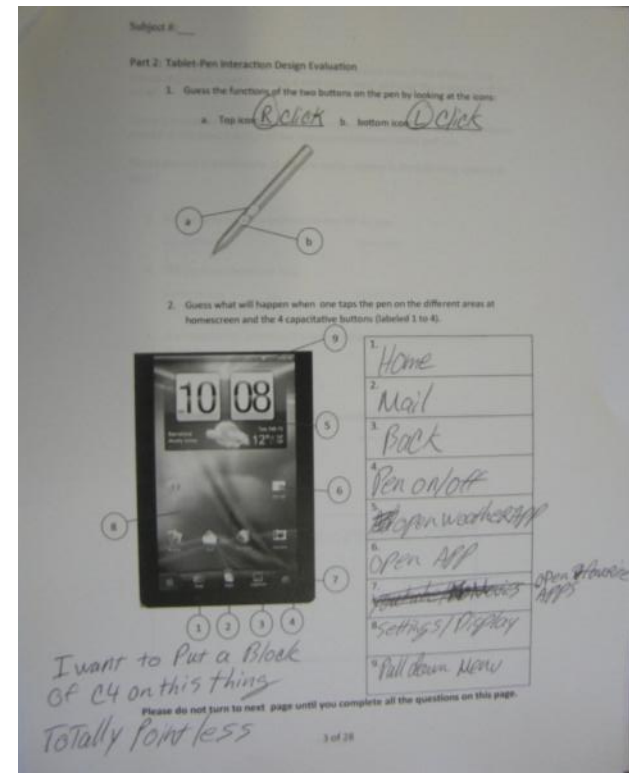
HTC Scribe/Flyer – Overall Pen/Tablet Product Experience

Usability and General Product Experience Issues

Insight:

Consumers are not likely to see compelling reasons of using pen/tablet as designed. It is a niche product and may be useful for specific user segments such as students and professional designers.

	Observations	Comments
Context of Use	Most respondents questioned whether the pen/tablet will be needed or used in the way as designed. The combination may be more useful for specific user segments such as students or designers.	Average consumers may not have a compelling need to use the feature-rich apps enabled by pen. It may be niche product in the market.
Interaction Design	Most respondents had negative remarks and experience with the interaction design paradigm of pen/tablet .	Consumers expect the pen will function as extension of finger touch. Changing the mode of the pen from finger touch to drawing should be done through the pen, but not using a dedicated button on the tablet.
Visual Design	Visual design of the Pen Options Palette was acceptable and reasonably well-designed.	The color palette icon can be improved by adding tints of color. The design can also benefit from bigger and meaningful icons.
Capability & Information Architecture	Some feature enhancements were suggested (e.g., mixing colors, XML-based pen edits)	Mixing color can be designed in a deeper menu hierarchy. Cross-platform compatibility and more seamless integration with existing apps were desired.



Pricing

Tablet & Pen Combination

PRODUCT CONCEPT EVALUATION

HTC Scribe/Flyer – Product Concept Statements

Product Concept statement from htc.com (July 6, 2011)

Product Overview

A good idea can take you anywhere. So we made a tablet that does the same.

The pen is mightier

Fact: Your finger is not a pen. So why do so many tablets treat it as one? On this tablet with digital pen accessory, you're able to take notes that sync with a recording of your meeting, highlight an important passage of an e-book or document, and even doodle on a picture or create your own.

Sync your ink

Capturing ideas is great. Being able to access and search those ideas from anywhere is way better. With this tablet, all of your notes and annotations are instantly saved to your Evernote account.

Your never-ending journal

With an expandable memory slot, you'll run out of ideas before you run out of places to keep them.

Life on the go

You know life happens on both sides of the screen.

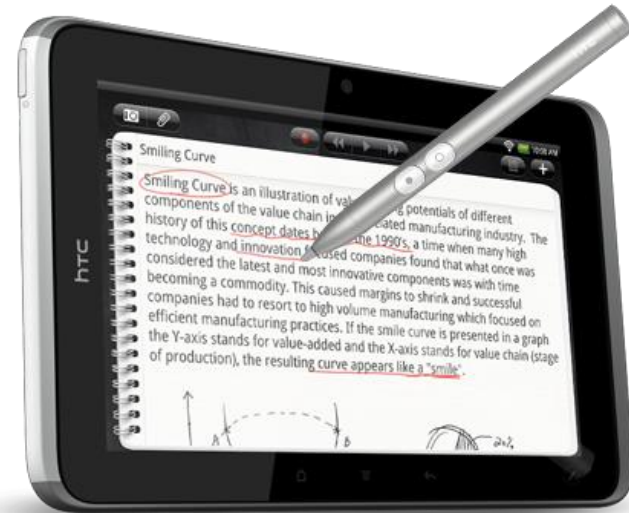
That's why we built a 5 MP camera into the back of the tablet for capturing your vision the world in HD.

And a front-facing camera for capturing your expressions in video chat.

Small. But not small

At 7", it's small enough to fit in your pocket, but with Android 2.3, that has been optimized for your tablet, and a 1.5GHz Qualcomm® Snapdragon® processor, it won't be spending much time there.

Price: \$499 (Tablet + Pen)



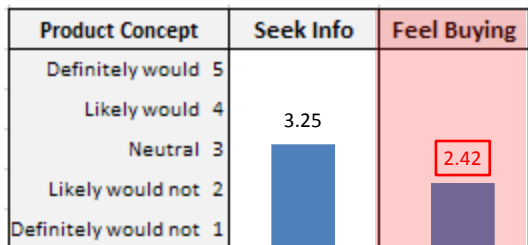
HTC Scribe/Flyer – Product Concept Scales

Product Concept Ratings

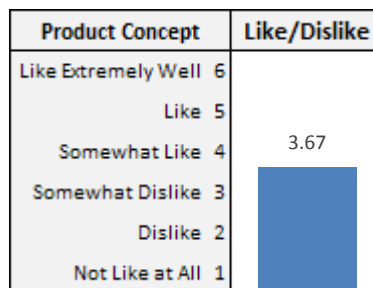
Insight:

Consumers generally have low interest in purchasing HTC Flyer and Scribe. The products (tablet with pen) fail to deliver adequate value to the consumers and do not meet customer expectations and needs.

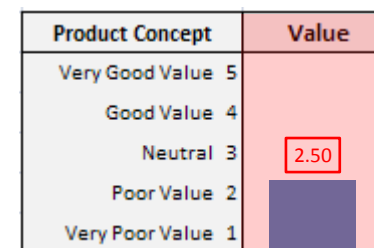
1. How likely you would seek more information about the product?
2. How do you feel about buying the product?



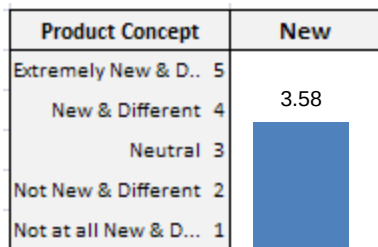
3. How much do you like the product?



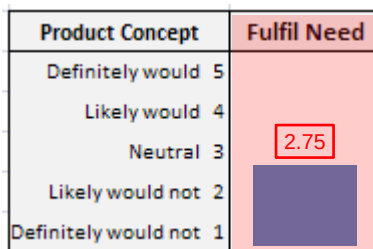
4. Considering the price, how do you feel about the value of the product?



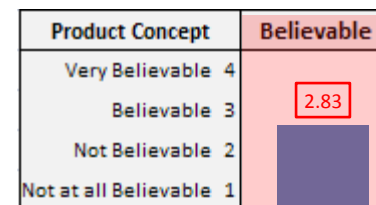
5. How would you rate the product in terms of being new and different from other products currently available?



6. How well would the product solve a problem or fulfil a need for you?



7. Which statement best describes how you feel about the believability of the statements made about the product?



HTC Scribe/Flyer – Product Concept Scales

Product Concept Ratings Distributions

Insight:
Consumers generally have low interest in purchasing HTC Flyer and Scribe. The products (tablet with pen) fail to deliver adequate value to the consumers and do not meet customer expectations and needs.

- How likely you would seek more information about the product?
- How do you feel about buying the product?

Product Concept	Seek Info	Feel Buying
Definitely would	5	2
Likely would	4	5
Neutral	3	2
Likely would not	2	0
Definitely would not	1	3

- How would you rate the product in terms of being new and different from other products currently available?

Product Concept	New
Extremely New & D..	5
New & Different	4
Neutral	3
Not New & Different	2
Not at all New & D...	1

- How much do you like the product?

Product Concept	Like/Dislike
Like Extremely Well	6
Like	5
Somewhat Like	4
Somewhat Dislike	3
Dislike	2
Not Like at All	1

- How well would the product solve a problem or fulfil a need for you?

Product Concept	Fulfil Need
Definitely would	5
Likely would	4
Neutral	3
Likely would not	2
Definitely would not	1

- Considering the price, how do you feel about the value of the product?

Product Concept	Value
Very Good Value	5
Good Value	4
Neutral	3
Poor Value	2
Very Poor Value	1

- Which statement best describes how you feel about the believability of the statements made about the product?

Product Concept	Believable
Very Believable	4
Believable	3
Not Believable	2
Not at all Believable	1

 Distribution of number of respondents (total N=12)

Van Westendorp Pricing

HTC Flyer Tablet with Scribe

Insight:

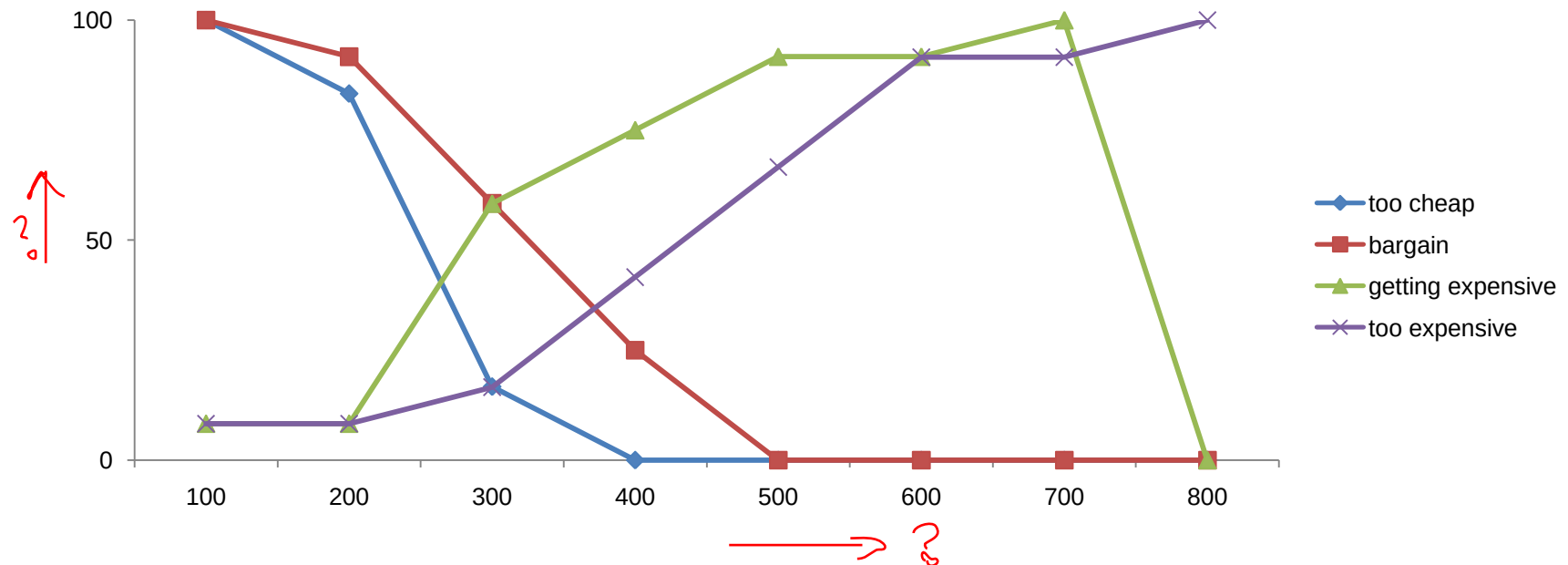
The current retail price of HTC Flyer Tablet and Scribe is far too high to be attractive to consumers. With a broken tablet/pen product experience, this tablet is unlikely to succeed in the market.

The range of acceptable prices of HTC Flyer Tablet with Scribe was between \$260 and \$375 (N=12)

Point of marginal cheapness = \$260 (approximate)

Point of marginal expensiveness = \$375 (approximate)

(Retail: \$499.99 as of July 6, 2011)



retail at \$499.99 with Scribe bundled in early July 2011 but Scribe is not bundled as of the date this report is published

Participant Demographics

APPENDIX

Participants Demographics

Smartphone owners – Tablet owners or intenders

Name	Age	Gender	Mobile Phone	Tablet
Colin M	44	M	LG Optimus	Motorola XOOM
Amy N	25	F	HTC Touch 4G	
Diane B	44	F	Motorola Droid X	
Joseph F	29	M	Motorola Defy	
Dana C	30	F	HTC Droid Incredible	
Jared P	35	M	Apple iPhone 4	Apple iPad
Natalie P	48	F	Samsung Transform	Samsung Galaxy Tab 7"
Joaquin A	25	M	Samsung Vibrant	
Oscar G	38	M	Apple iPhone 4	
Valerie L	27	F	Apple iPhone 3GS	
Stephen H	31	M	Apple iPhone 4	
Grant K	44	M	Samsung Captivate	Apple iPad 2
Chad B	40	M	Motorola Droid X	

Sang Hung, Nibha Jain, Sarena Yang
User/Product Research
Design & Technology Lab
Samsung Telecommunications America

Stephanie Vance
Strategic Marketing Research
Samsung Telecommunications America